SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
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Date:	06/10/2026	**Duration:**	60 minutes

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Annexure A – Case Study Analysis

Instructions to Students:

Each student will be allotted **one problem statement**. Analyze the assigned problem systematically by following the steps given below. Your report should demonstrate your understanding of software engineering concepts and your ability to design an appropriate software solution.

Based on your allotted problem statement, prepare a report by answering the following questions:

1. **Understand the Problem Statement**
   - Explain the problem in your own words.
   - Identify the objectives of the proposed system.
   - State the scope of the problem.
2. **Identify Key Stakeholders and Challenges**
   - Identify all the stakeholders involved in the system.
   - Explain the roles and responsibilities of each stakeholder.
   - Discuss the major challenges and issues faced by the stakeholders.
3. **Analyze the Current Approach**
   - Describe the existing system or current process.
   - Identify its strengths and limitations.
   - Analyze the problems or gaps that need to be addressed.
4. **Propose a Suitable Solution Using Software Engineering Principles**
   - Design a software-based solution for the given problem.
   - Describe the major functional modules of the proposed system.
   - Identify the functional and non-functional requirements.
   - Explain how software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security are incorporated into your solution.
5. **Justify the Chosen Methodology/Framework**
   - Select an appropriate Software Development Life Cycle (SDLC) model or software development framework (e.g., Agile, Scrum, Waterfall, Spiral, DevOps).

- Justify why the selected methodology/framework is suitable for the proposed system.
- Explain how it supports successful project development and maintenance.

6. **Discuss Expected Outcomes and Limitations**
   - Describe the expected benefits of the proposed solution.
   - Explain how the solution addresses the identified challenges.
   - Discuss potential risks, implementation challenges, and limitations.
   - Suggest possible future enhancements.

PROBLEM STATEMENT:

Vehicle Service and Maintenance Management System: Develop a system that allows vehicle owners to book service appointments, maintain service history, receive maintenance reminders, and track service status.

Vehicle Service and Maintenance Management System

1. Understand the Problem Statement

Problem Explanation

Vehicle owners often face difficulties in booking service appointments, keeping track of previous services, remembering upcoming maintenance, and knowing the current status of their vehicle service. Many service centers still depend on phone calls, manual records, or paper-based processes.

The Vehicle Service and Maintenance Management System is proposed to provide a centralized software platform where vehicle owners can book service appointments, view service history, receive maintenance reminders, and track the status of their vehicle service.

Objectives

- To provide easy online service appointment booking.
- To maintain complete vehicle service history digitally.
- To send reminders for upcoming maintenance and servicing.
- To allow customers to track the current service status.
- To reduce manual work for service centers.
- To improve communication between customers and service staff.
- To provide better organization and management of service records.

Scope

The system covers:

- Customer registration and login.
- Vehicle registration and management.
- Service appointment booking.
- Service history management.
- Maintenance reminders.

- Service status tracking.
- Service center/admin management.
- Notifications to customers.
- Basic service and billing information.

The system can be used by individual vehicle owners, service advisors, mechanics, and service center administrators.

2. Key Stakeholders and Challenges

Stakeholder	Role & Responsibility	Major Challenges
Vehicle Owner	Books appointments, views history, receives reminders and tracks service	Forgetting service dates, difficulty tracking service progress
Service Advisor	Manages appointments and communicates with customers	Handling multiple bookings manually
Mechanic	Performs inspection and repair and updates service status	Difficulty accessing accurate vehicle/service information
Service Center Admin	Manages users, services, staff and records	Managing large amounts of data
System Administrator	Maintains the software, security and database	Security, system maintenance and reliability

Major Challenges

- Manual appointment management can cause scheduling conflicts.
- Paper records can be lost or damaged.
- Customers may forget regular maintenance dates.
- Customers may not know whether their vehicle is being inspected, repaired, or is ready.
- Service staff may find it difficult to manage multiple vehicles simultaneously.
- Maintaining accurate and secure service records can be challenging.

3. Analyze the Current Approach

Existing System / Current Process

In a traditional system, the vehicle owner usually contacts the service center through phone calls or direct visits to book an appointment. The service center records customer and vehicle details manually or in separate systems.

The vehicle is then inspected and repaired by the mechanic. Customers may need to contact the service center again to know the service progress. Previous service records may be stored in paper files or scattered digital records.

Strengths

- Simple and familiar process.
- Does not require advanced technical knowledge.
- Suitable for small service centers.
- Direct communication between customer and service staff.

Limitations

- Appointment booking can be time-consuming.
- Possibility of scheduling conflicts.
- Manual records can be lost or damaged.
- No automatic maintenance reminders.
- Difficult for customers to track service status.
- Searching old service records takes time.
- Increased workload for service center staff.

Problems / Gaps

The major gap is the lack of a centralized digital platform connecting vehicle owners and service centers. The proposed system addresses this by providing appointment booking, service history, reminders, and real-time service status in one system.

4. Proposed Software Solution

Proposed System

A web/mobile-based Vehicle Service and Maintenance Management System will be developed to digitally manage the complete vehicle servicing process.

The customer can register their vehicle, select a service, choose an available appointment slot, and receive confirmation. During servicing, the service center can update the vehicle's status such as:

Booked → Vehicle Received → Inspection → Service in Progress → Service Completed → Ready for Pickup

The customer can view these updates through the system.

Major Functional Modules

1. User Management

- Registration and login
- Customer profile management
- Role-based access

2. Vehicle Management

- Add vehicle details
- Vehicle number/model information
- Service records for each vehicle

3. Appointment Management

- Select service type
- Select date and time
- Check available slots
- Book/cancel appointments

4. Service History

- Previous service details
- Repairs performed
- Parts replaced
- Service date and cost

5. Maintenance Reminder

- Periodic service reminders
- Oil change reminders
- Other maintenance notifications

6. Service Status Tracking

- Track vehicle service progress
- Status updates from service center
- Customer notifications

7. Admin/Service Center Module

- Manage appointments
- Assign mechanics
- Update service status
- Manage service records
- Manage customers and vehicles

Functional Requirements

- User registration and login.
- Vehicle registration.
- Appointment booking and cancellation.
- Service history storage and retrieval.
- Maintenance reminder generation.
- Service status updating.
- Customer notification.
- Admin and mechanic management.

Non-Functional Requirements

- **Security:** Protect customer and vehicle information.
- **Usability:** Simple and user-friendly interface.
- **Reliability:** System should work consistently without data loss.
- **Scalability:** Support increasing numbers of users and vehicles.
- **Maintainability:** Easy to modify and update.
- **Performance:** Appointment and service information should load quickly.

Software Engineering Principles

Modularity:
The system is divided into independent modules such as booking, vehicle management, reminders, and service tracking.

Scalability:
The system can support more customers, vehicles, and service centers as usage increases.

Maintainability:
Changes to one module can be made without significantly affecting other modules.

Security:
Authentication and role-based access control can restrict users to authorized functions.

Reliability:
Service information is stored digitally in a centralized database, reducing the risk of losing physical records.

Usability:
A simple interface allows customers to easily book services and check their vehicle status.

5. Justification of Methodology / Framework

Selected Methodology: Agile – Scrum

Agile Scrum is suitable for developing the Vehicle Service and Maintenance Management System because the system contains multiple modules that can be developed and tested incrementally.

The project can be divided into different sprints:

Sprint	Main Activities
Sprint 1	User registration and login
Sprint 2	Vehicle and profile management
Sprint 3	Appointment booking
Sprint 4	Service history and reminders
Sprint 5	Service status tracking
Sprint 6	Testing, security and final integration

Why Agile Scrum?

- Allows development in small increments.
- Customer feedback can be incorporated after every sprint.
- Problems can be identified and corrected early.
- Different modules can be tested independently.
- Requirements can be changed easily.
- Continuous testing improves software quality.

This is especially appropriate because the assessment itself emphasizes applying agile project management techniques and selecting a suitable SDLC/framework.

6. Expected Outcomes and Limitations

Expected Outcomes

The proposed system is expected to:

- Reduce manual appointment management.
- Make service booking faster and easier.
- Maintain organized digital service history.
- Help customers remember maintenance schedules.
- Provide better visibility of service progress.
- Reduce communication gaps between customers and service centers.
- Improve service center efficiency.
- Provide better customer satisfaction.

Risks / Implementation Challenges

- Initial development and implementation cost.
- Data security and privacy risks.
- Users may require time to adapt to the new system.
- Internet connectivity may affect access.
- Incorrect information entered by staff can affect service records.
- System maintenance and database management are required.

Limitations

- The system depends on internet/network availability.
- Automatic reminders are useful only when correct vehicle/service data is available.
- Real-time status tracking depends on service staff updating the system.
- Advanced vehicle diagnostics may require integration with external vehicle diagnostic systems.

Future Enhancements

- AI-based predictive maintenance to predict possible vehicle problems.
- Integration with IoT/vehicle sensors for real-time vehicle health monitoring.
- Online payment and digital invoice generation.
- GPS-based service center search.
- Mobile application for Android and iOS.
- AI chatbot for customer support.
- Spare-parts inventory management.
- Analytics dashboard for service centers.

Conclusion

The Vehicle Service and Maintenance Management System provides a centralized digital solution for managing vehicle servicing activities. It simplifies appointment booking, maintains service history, provides maintenance reminders, and enables service status tracking. By applying software engineering principles such as modularity, scalability, security, reliability, and maintainability, the system can provide an efficient and user-friendly solution for both vehicle owners and service centers. Agile Scrum is suitable because the system can be developed incrementally through multiple sprints with continuous feedback and testing.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Understanding (30%)	Clearly explains the problem, objectives, and scope with complete understanding.	Explains the problem with minor omissions.	Basic understanding with limited explanation.	Poor understanding; objectives and scope are unclear or missing.
Stakeholder & Challenge Analysis (25%)	Identifies all stakeholders, their roles, and challenges with clear analysis.	Identifies most stakeholders and challenges with adequate analysis.	Identifies only a few stakeholders or challenges.	Stakeholders and system analysis are incomplete or inaccurate.
Current System Analysis (20%)	Thoroughly analyzes the existing system, strengths, weaknesses, and identifies all gaps.	Good analysis with most strengths and weaknesses identified.	Limited analysis with partial identification of issues.	Proposed solution is unclear or inappropriate; requirements are largely missing.
Proposed Software Solution & Software Engineering Principles (15%)	Proposes an innovative, feasible solution applying appropriate software engineering principles (modularity, scalability,	Suitable solution with good application of software engineering principles.	Basic solution with limited application of software engineering principles.	Methodology is inappropriate, poorly justified, or not discussed.

	maintainability, security, etc.).			
Methodology/ Framework Justification (10%)	Selects the most suitable SDLC/model/framework and provides strong technical justification.	Suitable methodology selected with adequate justification.	Methodology selected but justification is weak.	Outcomes, limitations, or future enhancements are missing; report lacks organization and clarity.

Criteria	**Mark**
Problem Understanding (30%)	/5
Stakeholder & Challenge Analysis (25%)	/5
Current System Analysis (20%)	/5
Proposed Software Solution & Software Engineering Principles (15%)	/5
Methodology/ Framework Justification (10%)	/5
TOTAL	/25